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On the ethics of utilizing prediction markets for forecasting hydrological catastrophe

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As peer-to-peer prediction markets increasingly allow real-money wagering on environmental events, public scientific agencies are being drawn into the role of financial "settlement oracles," the designated sources of truth against which contracts resolve. The purpose of the present paper is to examine the ethical and operational risks that follow when AI-accelerated hydrologic forecasting becomes the settlement basis for such markets, with attention to flood and drought management. We develop a taxonomy of settlement oracles that distinguishes physical observations, certified records, model forecasts, official warnings, and administrative classifications, and we reframe the parallel between forecasting and market workflows as a comparison of partially analogous information systems whose prices are not calibrated probabilities. Three cases selected for maximum variation ground the analysis: a real-money contract settling on the observed crest of a fatal river flood, a federal prosecution alleging trading on classified government information, and a contract settling on a federal drought classification. Additionally, a stylized incentive model shows that the expected profit from manipulating a settlement value grows with market liquidity, and thus that thin markets deter manipulation only by accident. Applying consequentialist, deontological, and noxious-market frameworks across five modes of professional engagement, we conclude that trading by forecasters on products they produce or can influence is unethical without qualification, that agency use of market signals is at most conditionally permissible, and that the audit benefits claimed for these markets can be captured through sanctioned adversarial testing. We close with recommendations for agencies, professional societies, regulators, platforms, and affected communities.

KEYWORDS
artificial intelligence, forecasting, hydrology, prediction markets, water resources

1 Introduction

Prediction markets have a long and contested record as tools for collective forecasting. Advocates of these markets point to the wisdom of crowds and to instances in which market prices have outperformed expert polling (Wolfers and Zitzewitz, 2004; Cutting et al., 2025). Stakeholders of platforms such as Polymarket (i.e., users, employees and vested individuals) argue that economic incentives push prices toward true odds as more knowledgeable participants enter the market (Ahart, 2026). Indeed, a serious research literature supports parts of this claim: Hanson (2003) developed

the design theory for markets intended to inform decisions, Wolfers and Zitzewitz (2004) documented settings in which prices matched or beat expert judgment, and Arrow et al. (2008) argued for regulatory room to operate such markets precisely because of their forecasting value. Skeptics, however, note that markets reward profit rather than truth, and that the conditions which make an event tradable do not transfer cleanly to scientific forecasting. Indeed, the finance scholar Richard Borghesi has argued that prediction markets can be useful supplements for gauging how the public receives scientific information, but that they are not substitutes for models, peer review, or expert judgment, and that they grow less informative as the traders setting prices lose subject expertise (Ahart, 2026). In other words, the participants who set a market price are often not the participants who hold the relevant scientific knowledge. This dynamic has been described as a form of epistemic stratification, in which technical elites audit the underlying data while the public consumes a sanitized, capital–driven signal (Rohanifar et al., 2026). Water forecasting occupies a particularly consequential position within this debate, because the outcomes it predicts carry implications for people and the environment that the events these markets typically price do not. Our argument leaves the accuracy findings of the market design literature intact and concerns instead what happens at the settlement layer when the events being priced are hydrologic disasters and the records they settle against are public scientific products. In other words, we grant for the sake of argument that markets can forecast, and we ask what happens to the science when official products become the instruments on which bets settle.

The purpose of the present paper is to examine whether it is ethical to forecast water resources for the purpose of gaming a prediction market, and what obligations follow for the professionals and agencies whose products these markets consume. The insurance industry has priced flood and drought risk for many decades, and it has framed this activity as a public service even as it profits from the spread between premiums and payouts. Peer-to-peer prediction platforms, however, strip away that framing. For example, Kalshi and comparable venues allow participants to take positions on resolvable outcomes, to settle those positions against an authoritative public record, and to receive payment in near–real–time (Diercks et al., 2026). In this way, water management decisions, as well as the data and forecasts that inform them, may be priced on these platforms whether or not the agencies producing the underlying science intend it. This involuntary conscription of public science into a financial settlement mechanism creates a systemic vulnerability where markets incentivize predicting the agency's official report rather than the physical environment itself.

The evidence for this concern is partly documented and partly prospective, and the two should be kept separate. Real-money contracts that settle against federal observational products already exist, and they are no longer confined to weather. Platforms list precipitation and temperature contracts that resolve against named National Oceanic and Atmospheric Administration (NOAA) records. Additionally, in August 2026, while record flooding was still moving through central Indiana, Polymarket listed a set of contracts settling on the peak gage height recorded at a named U.S. Geological Survey (USGS) streamgage. Section 5.1 quotes the operative rule of that contract and documents its terms. Polymarket has additionally listed a set of contracts settling on the U.S. Drought Monitor exceptional drought classification, which Section 5.3 documents. What does not yet exist is a contract settling on a model-generated forecast or an official warning, such as a river stage forecast or a flood watch. Classes 3 and 4 of the taxonomy developed in Section 4 remain prospective; classes 2 and 5 no longer are. This distinction organizes much of the analysis that follows, and the distance it marks is narrowing. The White River contract settles on an observation of the river rather than on an agency judgment about the river. However, the contract names the National Weather Service (NWS) flood stage as a reference point within its own text. In other words, an official warning threshold already appears inside a financial instrument, even though that threshold is not the value against which the instrument resolves, and the contract is therefore one step from the classes in which the settling value is itself an agency judgment. Part of our analysis is thus anticipatory, and we treat it openly as anticipatory ethics (Brey, 2012). We examine the exposure before it forms, for two reasons. The moment to examine an exposure of this kind is before platforms, participants, and settlement conventions harden around it, and the record of adjacent markets indicates that contract menus expand toward whatever an audience will trade. Where our evidence is prospective, we mark it as prospective and keep our claims conditional.

Flood–risk probabilities supply a strong prior for the dynamic probability of an individual flood event, which is precisely the input that a prediction market requires. Additionally, technology companies hold the computational resources and the artificial intelligence expertise to generate probabilistic forecasts at a scale that public agencies and individual traders cannot match, and thus the potential to dominate such a market (Lopez-Gomez et al., 2026). In this setting, artificial intelligence occupies a space between legitimate risk prediction and unethical exploitation through opaque, manipulable, or under–supervised systems (Binesh and Ghaharian, 2025). A key ethical question is how values and market-driven AI prediction–based water decisions come into play for consequential hydrologic disasters. With considerations of data consent, bias, and equitable allocation, if the prediction market can lead to a reliable, potentially life-saving outcome, is it ethical to let it, or prevent it? We address this question with the oracle paradox in Section 4.1. Technology enabling this scenario is rapidly building, namely a financial market that settles on the forecasts themselves. We treat this layer as a management–decision problem rather than as a question of market integrity alone. In other words, when a forecast acquires financial value, the incentive to profit from that forecast can bend the operational decisions the forecast informs, such as flood warnings, reservoir operations, and drought declarations. The cost of that distortion then falls on communities who never see it happen.

2 Approach

This paper is a conceptual analysis, and this section makes its method explicit. The argument draws on three literatures: the practice of probabilistic hydrologic forecasting (Duan et al.,

2019), the economics of prediction markets and their interpretation (Hanson, 2003; Wolfers and Zitzewitz, 2004, 2006; Manski, 2006; Arrow et al., 2008; Dubach, 2026), and applied ethics, in particular consequentialist and deontological analysis and Satz's (Satz, 2010) account of noxious markets. Sources were selected to pair foundational work in each literature with recent scholarship and reporting on market behavior, and we cite journalism and government records only for matters of documented public fact.

Three ethical frameworks structure the normative analysis. They were chosen because they generate distinct and sometimes opposing verdicts on the same conduct, which forces the analysis to state where its conclusions depend on framework choice and where they do not. The frameworks are applied to defined modes of professional engagement (Section 10.1) rather than to participation in general, because the ethics of a forecaster trading on pre-release access differ from the ethics of an agency reading a market price as one signal among many.

Additionally, the empirical grounding follows a structured comparative case design. Three cases were selected for maximum variation (Flyvbjerg, 2006) along the dimension that matters most to the argument, the nature of the settlement oracle: an observation-settled flood-crest contract listed on a public platform while the flood it priced was in progress (Section 5.1); a federal prosecution alleging trading on material nonpublic government information (Section 5.2); and a contract settling on a federal drought classification (Section 5.3). All three are documented listings or proceedings rather than constructed scenarios. The limitations of this design are stated here and observed throughout. The paper contains no original data, and the stylized model in Section 8 illustrates a hypothetical incentive structure rather than measuring one. A market settling on model-generated forecasts or official warnings has not yet formed, so claims about such a market are conditional. Anticipatory analysis carries symmetric risks of overstatement and understatement (Brey, 2012), and we have tried to manage the first risk with conditional language and the second by grounding mechanisms in documented cases from adjacent domains.

3 Hydrologic forecasting and prediction markets as partially analogous information systems

Best practices for probabilistic forecasting of hydrologic systems are well established (Duan et al., 2019). Table 1 sets out this standard operational sequence: specifying the event, assimilating available observations, generating predictions, calibrating the output into an event probability, and verifying the product against the observed outcome. The table also includes partially analogous stages of prediction-market participation, as well as non-equivalences, confounders and failure modes. The hydrologic forecast represents uncertainty and is updated through time as new information arrives.

Artificial intelligence is changing how these predictions are made, allowing engineers to produce probabilistic forecasts using procedures that directly mirror, or surpass, the traditional data-assimilation workflow. Artificial intelligence-based hydro models learn model parameters from data directly, replacing explicit prior distributions, latent or stochastic sampling replaces ensemble perturbation, and fine-tuning replaces hydrologic post-processing. Uncertainty can be built into these systems in several ways, such as through a mixture model embedded in the forecasting architecture (Klotz et al., 2022) or through training a model directly on the bounds of uncertainty (Nourani et al., 2022). These methods alter the route to the forecast rather than the operational object required at the end of it. Both AI-based and physically based approaches can produce a predictive distribution for a hydrologic variable and an event probability, $p_{H,t_i} = P(A \mid \widetilde{Y}_{t_i})$. Here, $P(\cdot)$ denotes probability, A is the specified event, t_i is the forecast issue time, and $\widetilde{Y}_{t_i}$ is the information available at that time. This probability may inform the valuation of a contract on A, but it is not itself a market price.

A prediction market is a peer-to-peer trading structure in which the price of a binary contract is often read as a continuously updating probability. Participants may begin with beliefs informed by climatology, official guidance, or expert judgment; observe public information; revise their beliefs as forecasts and observations arrive; and express those beliefs as trades. The order book then produces bid and ask quotes, executed transactions, and price histories. These observables may respond to new information, but they also reflect trading incentives and constraints. When the event resolves, the contract settles according to its rules and designated oracle. Settlement is distinct from forecast verification: the resolution of one binary contract does not establish that its preceding prices were calibrated probabilities.

Reading a market price as a calibrated probability requires assumptions that real markets often violate. Manski (2006) showed that the price of a binary contract equals the mean belief of participants only under restrictive conditions and can diverge substantially from any well-defined aggregate belief under plausible heterogeneity. Wolfers and Zitzewitz (2006) identified conditions under which prices approximate mean beliefs, as well as the effects of risk aversion, wealth and collateral constraints, and trading frictions. Bid–ask spreads, thin liquidity, strategic trading, fees, and settlement frictions further separate price from probability, especially for low-probability contracts of the kind associated with hydrologic disasters (Dubach, 2026).

At minimum, four market objects must be distinguished: contemporaneous bid and ask quotes, an executed transaction price, a volume-weighted average price over a specified window, and a statistically adjusted implied probability. Only the last is a candidate for direct comparison with a calibrated hydrologic probability, and its construction requires assumptions about market microstructure. An execution price can nevertheless enter an informed actor's expected-payoff calculation. Equation 1 shows a contract paying \$1 if A occurs,

$$\Delta_{t_i} = P(A \mid \widetilde{Y}_{t_i}, I) - m^{\mathrm{ex}}_{t_i}, \tag{1}$$

where I is additional information available to the actor and $m^{\mathrm{ex}}_{t_i}$ is the price at which the actor can execute the trade. Thus, Δ_{t_i} is an expected per-contract payoff margin, not a claim that the execution price and the conditional probability are equivalent epistemic objects.

TABLE 1 Partially analogous information-processing workflows in hydrologic forecasting and prediction-market participation.

Phase	Hydrologic workflow	Hydrologic notation	Prediction-market analogue	Non-equivalences, confounders, and failure modes
1. Define the event	Specify event A, such as streamflow Q exceeding threshold q_A.	$A = \{Q > q_A\}$	Define a binary contract paying \$1 if its resolution criteria are met.	Contract criteria may differ from the physical event in timing, data source, or threshold.
2. Update information	Collect observations, forecasts, and model states at issue time t_i.	$\widetilde{Y}_{t_i}$, the information available at t_i	Participants observe public information; an informed actor observes $(\widetilde{Y}_{t_i}, I)$, where I is additional information.	Markets reflect only information expressed through trades and are affected by incentives and capital constraints.
3. Generate and aggregate	Generate and post-process an ensemble predictive distribution for Q.	$\left\{Q_{t_i}^{*(k)}\right\}_{k=1}^{K}$, $F^*_{Q,t_i}(q)$; $Q^{*(k)}$ is post-processed member k, K is ensemble size, and F^* is the calibrated predictive CDF.	Orders generate quotes and transactions through the order book.	Prices depend on capital, risk preferences, liquidity, market depth, and trading strategy.
4. Express probability and price	Convert the predictive distribution into the probability of A.	$p_{H,t_i} \equiv P(A \mid \widetilde{Y}_{t_i}) = 1 - F^*_{Q,t_i}(q_A)$	Distinguish bid b_{t_i}, ask a_{t_i}, transaction price $m_{t_i}^{\mathrm{tx}}$, volume-weighted average $\overline{m}_{W_i}$ over window W_i, and adjusted implied probability $\widehat{p}_{M,t_i}$, where M denotes the market. Only the last is comparable with p_{H,t_i}.	Price-to-probability adjustment depends on spreads, liquidity, risk preferences, constraints, fees, and settlement; calibration is not assured.
5. Update and act	Update the forecast as information arrives.	$\left\{P(A \mid \widetilde{Y}_{t_i})\right\}_{i=0}^{N}$, for $N+1$ forecast issue times	At execution price $m_{t_i}^{\mathrm{ex}}$, $p_{I,t_i} = P(A \mid \widetilde{Y}_{t_i}, I)$ is the actor's event probability and gives $\Delta_{t_i} = p_{I,t_i} - m_{t_i}^{\mathrm{ex}}$. For position S, λ is price impact per unit and $D = 1/\lambda$ is market depth; see Section 8.	Prices may move through hedging, liquidity demand, strategy, or price impact without new event information.
6. Verify and settle	Compare the forecast with the observed outcome.	Observed A or A^c, the complement of A	Settle against the contract's designated oracle; manipulation may target that process rather than $P(A)$.	Contract wording, oracle choice, delays, revisions, or manipulation may separate settlement from the physical event.

Hydrologic and price-impact notation follows Duan et al. (2019) and Kyle (1985); market limitations follow Manski (2006) and Wolfers and Zitzewitz (2006).

One well-known approach to trading a prediction market rests on position sizing rather than frequency. A participant balances exposure against the probability of being correct, keeps most capital protected from loss, and commits only a limited and survivable amount to those outcomes where the market appears to misprice the probability or the magnitude of the payoff (Taleb, 2007). The aim of this approach is the identification of rare opportunities where downside is bounded and upside is asymmetric, rather than frequent or aggressive betting (Taleb, 2012). In other words, this is a strategy built for tail risk, which is precisely the shape of flood and drought. Empirical evidence from decentralized markets supports this approach. For example, a persistent longshot spread premium rewards exactly this strategy, a finding that validates the use of advanced probabilistic forecasting to capture asymmetric upside on rare extreme events (Dubach, 2026). The edge therefore comes from holding a probability estimate that is sharper than the market's, which lets a participant trade either side of the misprice. A forecaster equipped with advanced probabilistic models can profit by buying a tail the market has underpriced or by fading one it has overpriced, and in both cases the advantage is the quality of the forecast rather than the direction of the bet.

A second feature of these markets warrants attention before the workflow is described. Both market prices and agency forecasts can harden into apparent fact through a loss of probabilistic detail, a process of knowledge scrubbing in which the uncertainty attached to a prediction fails to survive the communication chain to the people who use it (Rohanifar et al., 2026). This same erosion already occurs with ordinary hydrologic warnings. For example, a stated probability becomes, in public understanding, "the event will happen," and after several forecast events fail to materialize, it becomes "the event will not happen." A market price, presented as a single number, accelerates this flattening, because it reduces the distribution to one tradable figure and presents that figure with the authority of money.

3.1 What artificial intelligence changes

Most of the risks examined in this paper could arise, in principle, in a market populated by human traders using traditional models. Artificial intelligence does not create a new category of wrongdoing; rather, it changes five parameters of the existing categories, and each change widens the gap between the best-resourced participants and everyone else. The first is scale: machine learning systems now produce probabilistic forecasts across many catchments at low marginal cost (Lopez-Gomez et al., 2026), so a single participant can hold informed positions across an entire contract menu. The second is speed: learned models can update at the cadence of the market rather than the cadence of an operational forecast cycle. The third is opacity: a learned trading model resists the kind of audit that a physically based model invites, so detecting whether a participant is forecasting the environment or the agency becomes harder from the outside (Binesh and Ghaharian, 2025). The fourth is automation: agents can monitor order books and execute continuously without human attention, which matters most in thin markets where a small number of automated participants set the price. The fifth is concentration: the computational resources behind these systems sit with a small set of companies, and market-making liquidity already concentrates among a small tail of well-resourced participants (Dubach, 2026). Where the following sections identify a manipulation path or an information asymmetry, these five parameters determine how wide it opens. In this way, artificial intelligence amplifies existing asymmetries rather than creating new ones.

4 A taxonomy of settlement oracles

A settlement oracle is the designated source of truth against which a contract resolves. Commentary on prediction markets tends to treat all settlement sources as a single class. However, a contract that settles on a measured rainfall total is a different object from one that settles on a flood warning or a drought classification, because the underlying sources differ in who produces them, how much judgment they embed, how they can be manipulated, and what depends on them downstream. Table 2 distinguishes five classes of oracle and records, for each, the principal actors, the information asymmetries those actors hold, the paths by which the settling value could be manipulated, the safeguards that currently exist, and the consequences if the oracle is gamed.

Two properties organize the table. Verifiability declines from the top of the table to the bottom, because a station precipitation total is close to a direct physical measurement, while a drought category is an expert synthesis of indicators and local reports. Embedded judgment rises as verifiability falls, and with it both the surface available to manipulation and the difficulty of detecting that manipulation. Flood crests occupy an intermediate position. A peak flow is a physical quantity, but in most disaster scenarios it is reconstructed after the fact through routing assumptions, rating curves, and operational judgment rather than read directly off an instrument.

This gradient answers a reasonable objection to our argument. Observation-settled contracts of the kind quoted in Section 5.1 sit near the observational end of the table, and the existence of such contracts does not establish that operational forecasts are equally verifiable. We grant the point. Granting it, however, does not protect the products lower in the table, because the pressures acting on settlement convention push in the opposite direction. Markets prefer clean settlement, and physical measurement is hardest in precisely the low-probability hydrologic scenarios that disaster contracts would price. Settlement therefore drifts down the table, toward certified records, official products, and administrative classifications. In other words, the classes in which public trust is most exposed are also the classes toward which settlement convention moves.

4.1 The Oracle Paradox and Goodhart's Law

Whether the financialization of forecasts yields any scientific benefit depends entirely on what the contract settles against, a vulnerability we term the oracle paradox. If the oracle is an independent, physical observation, such as an automated gauge reading or a physically measured crest, then the market theoretically rewards forecasting that physical observation more accurately. However, in most hydrological disaster scenarios, decentralized markets cannot verify the physical environment directly, because quantities such as peak flow and flood water depth are hardest to measure in exactly the low-probability scenarios these contracts would price. Resolution against the official government record has therefore become the operative convention. If the oracle is a government-issued forecast or warning, the market rewards predicting the agency rather than the physical conditions. In this case, participants build surrogate models that match the agency's warning parameters whether or not the warning accurately reflects physical reality. This paradox leads to not an improved prediction about the event, but rather a mimicking of what is already known.

This dynamic is a direct manifestation of Goodhart's Law, which states that when a measure becomes a target, it ceases to be a good measure. By transforming a public safety forecast (the measure) into a financial settlement threshold (the target), the scientific utility of the forecast is compromised. The system settles into a pattern that tracks official statements rather than physical reality, generating no overall improvement in predictive accuracy while producing an unethical capitalization on disaster.

5 Three cases

The following cases were selected for maximum variation along the oracle dimension of Table 3, as described in Section 2. The first is documented and observation-settled, the second is documented and turns on information asymmetry rather than hydrology, and the third is documented and settles on an administrative classification.

TABLE 2 Taxonomy of settlement oracles for water-related event contracts.

Oracle class and example	Key actors	Information asymmetries	Manipulation paths	Existing safeguards	Consequences if gamed
1. Direct physical observation Automated gauge stage; station precipitation total	Instrument operators; telemetry providers	Physical access; knowledge of maintenance and outage schedules	Sensor tampering; spoofed or replayed telemetry	Quality-control flags; redundant sensors; automated range checks	Localized settlement error; corrupted observations may propagate into downstream models
2. Certified or quality-controlled record Monthly climate summary; finalized peak flow	Agency quality-control and records staff	Knowledge of pending revisions and release timing	Interference during the interval between preliminary and final issuance	Documented revision procedures; audit trails	Disputes over which figure governs settlement; erosion of trust in certification
3. Model-generated forecast River-stage forecast; seasonal streamflow outlook	Modelers; operational forecasters	Pre-release access to model runs, guidance, and post-processing decisions	Corrupting ingested inputs; tilting post-processing; trading ahead of public release	Reproducible pipelines; versioned inputs, models, and code	Forecasts tuned to the market rather than to operational users; Goodhart dynamics (Section 4.1)
4. Official warning or advisory Flood watch or warning; evacuation trigger	Warning coordinators; emergency managers	Control over issuance timing and decision thresholds	Delaying, accelerating, or suppressing release	Formal dissemination protocols; multi-office coordination	Distorted life-safety timing; the most severe direct-harm profile in the taxonomy
5. Administrative classification or declaration Drought category; disaster declaration	Interagency analysts; review committees	Knowledge of deliberations, pending designations, and administrative thresholds	Interfering with upstream data to move indicators; coordinated impact reporting; lobbying decision-makers	Multi-source convergence; published methodologies; interagency review	Relief and resource allocations redirected; statutory or administrative thresholds gamed

Verifiability declines and embedded judgment rises from Class 1 to 5.

5.1 An observation-settled contract: the White River flood crest at Indianapolis

In August 2026, record rainfall across central Indiana drove the White River past its 1913 benchmark at two upstream gages, ultimately resulting in a federal emergency declaration (Gersh, 2026). While that flood was still moving downstream, the decentralized platform Polymarket listed a real-money market on its crest, titled "What will the White River at Indianapolis peak at during the August 2026 floods?" The market opened on August 14, 2026 and split the outcome into four mutually exclusive bands, each listed and priced as a separate binary contract: under 14 feet, 14 to 17 feet, 17 to 20 feet, and 20 feet or above. Every leg named a federal scientific agency as its settlement oracle. The rule governing the highest band reads (Polymarket, 2026b):

"This market resolves "Yes" if the maximum gage height recorded at USGS streamgage 03353000 (White River at Indianapolis, IN) between market creation and 11:59 PM EDT August 31, 2026 is at or above 20.00 feet. Otherwise, this market resolves "No"."

The same rule then fixes which version of the agency record governs: "USGS real-time data is provisional and subject to revision; the last provisional value published by USGS for the specified window as of 11:59 PM EDT September 1, 2026 governs, regardless of any USGS revision, approval, or finalization published after that time." A further clause names the NWS flood stage at the location, 16.0 feet, as a reference point, and permits resolution on "a consensus of credible reporting citing USGS data" alongside the primary source. Settlement is executed on-chain through an optimistic oracle, and thus by a decentralized dispute mechanism, not by the agency named.

Read against Table 2, this contract settles on a class 2 oracle, and the case shows why the class matters. The physical quantity is directly observable, but the settling value is a figure in an agency record, and the contract goes further by deciding in advance which figure in that record will count. Indeed, by fixing settlement to the last provisional value published before a stated cutoff, and by disclaiming any later USGS revision, approval, or finalization, the contract writes the agency quality-control procedure out of the settlement. The consequence recorded for class 2 in the taxonomy is a dispute over which figure governs. Here that dispute has been resolved in advance in favor of the unverified number. In other words, a position pays out on a reading the agency has not yet checked, and a subsequent correction by the agency changes nothing. The fallback provision compounds this, because a consensus of credible reporting admits an undefined journalistic aggregate as a settlement input, and that fallback becomes available

in exactly the conditions a severe flood produces, when telemetry fails or publication lags.

Three features of the case carry forward into the analysis. The agency named in the contract had no role in creating or sanctioning it, which is mode 4 of Table 3 in its plainest form. The event priced was a disaster in progress and not a scheduled announcement, so the settling value was being generated by the same emergency the market was trading. The settlement source is an observation and not a forecast, which is the least exposed position in the taxonomy, yet the contract already reaches into the adjacent class by naming an NWS flood-stage threshold in its text. Regulated venues list comparable contracts resolving against named observation stations (Diercks et al., 2026). Thus the drift toward operational products that we treated as prospective has already begun at the observational end, and what remains prospective is settlement on the classes in which the value is itself an agency judgment.

5.2 An alleged insider case: United States v. Van Dyke

In April 2026, the Department of Justice unsealed a five-count indictment charging an active-duty U.S. Army soldier, Master Sergeant Gannon Ken Van Dyke, with unlawful use of confidential government information for personal gain, theft of non-public government information, commodities fraud, wire fraud, and making an unlawful monetary transaction. The indictment alleges that Van Dyke used classified information about the timing of a U.S. military operation in Venezuela to realize roughly $409,000, on an initial stake of about $33,000, across 13 Polymarket positions keyed to that operation's outcome (U.S. Department of Justice, 2026). The Commodity Futures Trading Commission filed a parallel civil complaint the same day.

The litigation is ongoing, and the procedural status recorded here is current as of August 26, 2026. Van Dyke pleaded not guilty on April 28, 2026. On June 8, 2026, the U.S. District Court for the Southern District of New York set a tentative trial date of December 7, 2026. On July 31, 2026, the defense moved to dismiss the indictment, arguing that Polymarket event contracts are not swaps within the meaning of the Commodity Exchange Act and that confidential military planning information is not property for purposes of the wire fraud statute; the government opposed that motion in August 2026, and it remained pending at this writing. The parallel civil action was stayed on August 10, 2026 pending resolution of the criminal case. Neither an indictment nor a civil complaint is a judicial determination, and the allegations remain untested at trial. Nothing in the analysis below turns on the outcome.

What the case documents, whatever its outcome, is the mechanism this paper turns on. First, privileged access to nonpublic government information is directly monetizable on these platforms, and the charging statutes, including commodities fraud, show that prosecutors regard event contracts as instruments to which insider principles apply. Second, detection ran through the platform: Polymarket identified the trading pattern, referred the matter to authorities, and afterward revised its rules to bar trading on misused confidential information. Third, position visibility cut both ways: on-chain activity made the trades observable, yet identifying coordinated or disguised activity on such platforms generally requires forensic analysis beyond the reach of most observers (Dubach, 2026). This case was selected because it is not hydrological: it isolates the information-asymmetry mechanism from any question about whether water markets exist yet. An operational forecaster's pre-release access to hydrologic guidance is, structurally, the same asset the indictment alleges was exploited.

5.3 An operational case: a D4 drought-classification contract

Polymarket has listed "Exceptional drought (D4) by state — week of August 25, 2026," a set of state-specific binary contracts settling against the U.S. Drought Monitor (USDM) (Polymarket, 2026a). For each listed state, the contract resolves "Yes" if D4, or Exceptional Drought, covers at least 0.05% of the state's land area in the USDM release for that week; otherwise it resolves "No." Settlement follows the first published release for the listed week; later revisions do not alter the result. This is a class 5 administrative classification rather than an observation-settled contract. The weekly USDM designation is an expert synthesis of indicators, observations, and locally submitted condition and impact reports. The live listing does not establish that any participant possesses nonpublic information or has attempted to influence the classification. It does, however, create a direct financial interest in the boundary drawn by that classification. Relevant information asymmetries could include pre-release knowledge of the deliberative process or knowledge of locally submitted reports not yet reflected in the published map.

The principal manipulation paths remain upstream. Interference with observations that inform drought indicators, or coordinated efforts to shape condition and impact reporting, could affect both the classification and the contract without tampering with the published USDM product. The settlement rule makes the timing consequential: a marginal classification near the 0.05% threshold, and the initial publication rather than a later revision, determines the financial outcome.

The downstream stakes make this more than a new weather-market category. Drought classifications can inform public drought responses and, in some programs, eligibility or priority decisions tied to category thresholds and duration. A contract on D4 therefore gives traders a financial interest in an administrative classification that may also guide relief and allocation decisions. The people affected by those decisions need not be market participants. Section 9 develops the consequences of an instrument built to organize drought response becoming a settlement oracle for speculation.

6 Insurance, risk-linked securities, and peer-to-peer markets

Insurance and prediction markets both price uncertain hydrologic outcomes, however they manage risk in opposite directions. Actuarial models pool long-horizon risk across

many policyholders in order to spread loss, and these models remain sensitive to the historical data on which they are built (Saunders et al., 2025). This sensitivity is a known fragility. For example, index–based flood insurance can swing materially with the choice of data source, and correlated catastrophe risk can trigger insolvency when losses cluster rather than diversify (Nayak et al., 2025). In this way, insurance carries deep uncertainty on its balance sheet by design and absorbs that uncertainty over time.

Prediction markets, by contrast, do the opposite. They settle short–horizon positions for immediate profit, and they carry no obligation to pool risk or to serve the insured. Much of the recent literature treats insurance as the default protection against flooding and proposes raising premiums in order to push more households into coverage (Bellia et al., 2025). Additionally, artificial intelligence is now being adopted for the predictive analytics that quantify that risk and the capital held against it (Pandiri, 2026). These same methods, in the hands of business interests with the means to dominate a market, transfer directly to prediction markets. In other words, the same artificial intelligence expertise that allows a startup to price flood and drought risk outside the traditional insurance system also allows that startup or individual to profit from decentralized, high–frequency markets in those same outcomes.

The boundary between insurance and event markets is a continuum rather than a wall, and the argument must concede as much. Catastrophe bonds transfer correlated disaster risk to capital markets and can pay out on parametric triggers (Cummins, 2008), weather derivatives settle on temperature and precipitation indices (Jewson and Brix, 2005), and index insurance pays on measured thresholds rather than adjusted losses (Kousky, 2019; Saunders et al., 2025). Short-horizon, parametric financial exposure to hydrologic outcomes therefore already exists, and a categorical distinction between these instruments and event contracts would not survive inspection. What peer-to-peer event markets change is position along four parameters. The first is participants and purpose: catastrophe bonds and derivatives are priced between institutions managing correlated exposure, while event contracts are open to any retail participant with no exposure to hedge. The second is the settlement object: parametric instruments settle on indices constructed for the purpose, while event contracts increasingly settle on operational public-safety products that exist to warn and to allocate. The third is granularity and immediacy: a catastrophe bond references a season and a region, while an event contract can reference one county and one date and settle within hours. The fourth is governance: regulated instruments carry disclosure and suitability regimes, while decentralized platforms carry neither. Satz's dimensions of vulnerability and harm then apply across the continuum rather than to a category label, so an instrument's standing tracks its position on these parameters: a catastrophe bond priced between institutions and a retail contract on whether a specific town floods sit at opposite ends of the same gradient.

The clustering that destabilizes insurance reappears as a perverse incentive within markets. Because priced risk concentrates tightly in particular geographies (Nayak et al., 2025), a water market may settle repeatedly on the same exposed region. This repetition can reward the building of infrastructure that protects a community against disaster, or it can reward the dismantling of that infrastructure, depending on which side of the contract a dominant participant holds. The market is indifferent to which outcome occurs, and this indifference is the ethical problem. In other words, the financially optimal position, and the public good, can point in opposite directions for the same community. From a noxious-markets perspective, the retail peer-to-peer end of this continuum is where the problems concentrate. Unlike traditional insurance, which theoretically attempts to pool and mitigate shared vulnerabilities, a prediction market settling on a catastrophic flood or drought relies on the desperate vulnerability of those communities. It transforms physical harm into a purely extractive, speculative vehicle, causing deep harm to the social fabric.

This raises a question that the research community has largely left implicit. Specifically, why should flood–risk pricing, whether conducted by an insurer or by a market maker, be treated as science when its purpose is to extract value rather than to forecast accurately? The two activities use the same probabilistic machinery and the same public data; however, they answer to different masters. Additionally, the peer–to–peer structure is not as democratic as it appears. Market-making liquidity tends to concentrate among a small tail of well-resourced participants, which allows technically advanced entities to dominate the ecosystem in much the same way that large insurers dominate their own (Dubach, 2026). In this way, the promise of a crowd–sourced forecast gives way to a small number of sophisticated actors who set the price.

There is reason to think that these markets measure something other than scientific probability. For example, when a hantavirus scare surfaced on a cruise ship in early 2026, a market on whether the World Health Organization would declare a pandemic spiked on the news shock, and then fell back as the alarm faded, even though the underlying scientific risk had not meaningfully changed. Indeed, the epidemiologist Bill Hanage observed that such bets reveal how the public feels more than what is likely to happen (Ahart, 2026). For water, this distinction may have consequences. A market on a flood crest may track public anxiety after a storm warning, rather than the probability that an operational model would assign, and a participant who understands the gap between sentiment and science can profit from that gap. However, markets are not uniformly detached from expertise. The climate scientist Zeke Hausfather has noted that some climate–related bets fall reasonably in line with expert estimates (Ahart, 2026). The performance of these markets is therefore uneven, which is the reason that treating any single market price as an authoritative forecast is hazardous.

7 Ways to game the system

Rapid commercialization can transform water forecasting from an objective scientific tool into a gamified financial instrument, and this transformation creates incentives to manipulate the underlying data, to exploit privileged information, and to distort the public record (Packin and Rabinovitz, 2026). Success within a market does not require an accurate forecast. Instead, it requires a profitable position, and there are several routes to such a position that do not pass through better science, shown in Figure 1 and described in detail below.

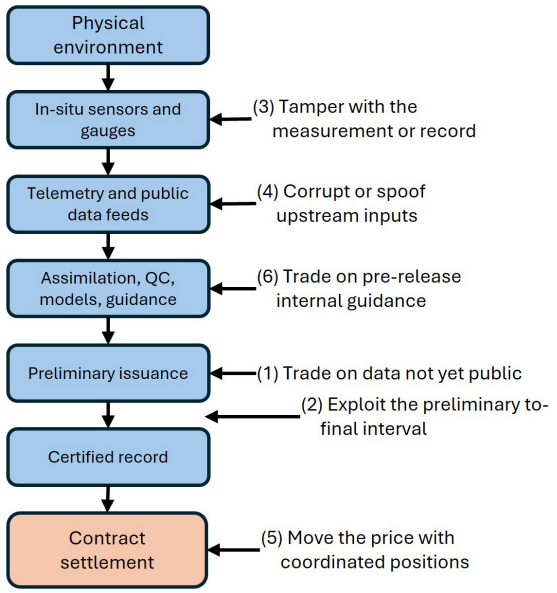


FIGURE 1
The operational hydrologic data pipeline and six routes to a profitable position that do not pass through better science. The routes are numbered as in the text, and each route is attached to the stage of the pipeline at which it operates.

The first is trading on data not yet public. An individual with advance sight of an observation or an official report can take a position before the rest of the market sees the same information. The second is exploiting the reporting pipeline directly, since the interval separating observation, preliminary issuance, and final certification is the window where advance knowledge or interference carries the most financial worth. The third is interference with the measurement or the record itself, manipulating the settling observation at an agency-operated station or gauge rather than the physical event. The fourth, and most concerning for water forecasting, is interference upstream in the data supply chain. Operational forecasting systems are not closed systems. They ingest precipitation records, numerical weather predictions, and *in-situ* water observations, which can be broadcast live to an on-line portal. So corrupted, spoofed, or selectively altered inputs could move a model output that settles a contract without any tampering at the final report. The fifth is ordinary market manipulation, the use of coordinated or outsized positions to move a price, to which thinly traded weather and water markets are especially susceptible. The sixth is insider trading on material nonpublic information, such as an employed operational forecaster taking a position based on routine access to internal hydrologic guidance before that forecast is published.

These concerns are not abstract. The Van Dyke prosecution (Section 5.2) documents that privileged access to nonpublic government information is directly monetizable on these platforms, and the pattern is not confined to a single case. In October 2025, Norwegian officials overseeing the Nobel Peace Prize examined a surge of bets favoring the eventual laureate that had been placed hours before the announcement (Ahart, 2026), a market movement consistent with someone trading on the outcome before it was public. Detection, however, is more difficult than these cases suggest. On decentralized platforms, public order-book feeds are poor indicators of true trade direction, and identifying coordinated manipulation therefore requires complex on–chain forensic analysis that most observers cannot perform (Dubach, 2026). The manipulation of a water forecast could therefore be both profitable and difficult to prove. Additionally, a comparable case involving an agency that issues hydrologic forecasts would damage more than the individual. It would damage the credibility of the institution and of the operational science that depends on public trust in impartial data. Viewed through a deontological perspective, this represents a categorical violation of professional ethics. A forecaster holds a fundamental duty to objectivity and public safety; exploiting the data pipeline treats the public's well-being and the agency's resources merely as a means to a financial end. The harm to a scientific agency is therefore worse than the harm to a betting exchange, because the agency's authority rests entirely on the perception that its data is clean and its duty to the truth remains uncorrupted.

7.1 What markets change relative to existing distortions

An objection from institutional realism deserves a direct answer here: operational agencies do not currently work free of incentive pressure, and a comparison that treats them as previously undistorted would overstate what markets introduce. Forecasters already weigh the cost of false alarms against the cost of missed events, and this tension is documented to shape warning decisions (Barnes et al., 2007). Drought and disaster declarations already sit inside a political economy in which relief budgets, agricultural constituencies, and electoral timing bear on where administrative lines are drawn. A version of the oracle paradox therefore predates financialization, because agencies have long known that their products are watched and that the watching carries consequences.

What a settlement market changes is the ownership, the immediacy, and the precision of the distortion. The first is ownership and direction: the existing pressures are diffuse and institutional, contested in public through hearings, appeals, and press coverage, whereas a market position creates a private, individualized, and largely invisible payoff that no hearing ever examines. The second is immediacy: institutional pressure operates at the cadence of budget and election cycles, whereas a contract settles at the cadence of the operational decision itself, within hours of a warning or a designation. The third is precision: political pressure bears on programs and regions, whereas a position can attach to one gauge, one threshold, one county, and one date. The claim is therefore comparative: markets concentrate distortion, attach a price to it, and privatize it, rather than introducing distortion into a previously incentive-free system. The safeguards of Section 12 therefore aim at the settlement layer rather than at institutional incentives in general.

8 A Stylized Model of the Manipulation Incentive

The incentive structure described above can be expressed using the familiar language of probabilistic hydrologic forecasting. Let A denote the hydrologic event on which a binary contract settles. For example, A may be the event that streamflow exceeds a threshold q_A. Following the probabilistic forecasting framework summarized in Table 1, the information available at forecast issue time t_i is denoted $\tilde{Y}_{t_i}$, and the post-processed hydrologic forecast distribution is $F^*_{Q,t_i}(q)$. The forecast probability of the event is shown below in Equation 2:

$$P(A \mid \tilde{Y}_{t_i}) = P(Q > q_A \mid \tilde{Y}_{t_i}) = 1 - F^*_{Q,t_i}(q_A). \tag{2}$$

This is the quantity familiar to a hydrologist: the probability assigned to an event given the observations, model states, meteorological forcings, ensemble forecasts, and other information available at t_i (Duan et al., 2019).

A prediction market represents the same event in a different way. Let $p_{M,t_i} \in [0,1]$ denote the current price of a binary contract that pays \$1 if A occurs and \$0 otherwise. The price can therefore be interpreted, subject to the usual qualifications about market efficiency and risk, as a market-implied probability of A. The operational hydrologic probability $P(A \mid \tilde{Y}_{t_i})$ and the market price p_{M,t_i} need not agree.

The remaining notation comes from the market rather than the hydrologic forecasting system. We adopt a reduced-form linear price-impact formulation motivated by Kyle (1985). Let S denote the number of contracts acquired and let λ denote the price change per unit position. Kyle's λ is therefore analogous to a market sensitivity: large λ denotes a thin market in which relatively little trading moves the price, whereas small λ denotes a deeper market. Equation 3 defines the market depth as

$$D = \frac{1}{\lambda}. \tag{3}$$

To connect this price-impact concept to the binary contract, Equation 4 shows the marginal execution price rises linearly as the position is accumulated,

$$p(s) = p_{M,t_i} + \lambda s. \tag{4}$$

Acquiring S contracts therefore moves the market from p_{M,t_i} to $p_{M,t_i} + \lambda S$. Equation 5 shows the linear impact on the execution price as:

$$\bar{p}(S) = p_{M,t_i} + \frac{\lambda S}{2}. \tag{5}$$

Equivalently, the total acquisition cost is the area under this linear price-impact curve, $p_{M,t_i}S + \lambda S^2/2$. This is the step that adapts Kyle's linear price-impact concept to the stylized execution of a prediction-market position.

Now suppose an actor possesses additional information I that is not reflected in the market price. In hydrologic terms, I might contain an unreleased observation, model run, ensemble forecast, data-assimilation update, or operational judgment. Equation 6 shows an actor conditioned on a larger information set than the public:

$$P(A \mid \tilde{Y}_{t_i}, I). \tag{6}$$

Equation 7 defines the actor's probability advantage over the market as

$$\Delta_{t_i} = P(A \mid \tilde{Y}_{t_i}, I) - p_{M,t_i}. \tag{7}$$

This quantity has a direct hydrologic interpretation. It is the difference between the probability of the settling event implied by the actor's information and the probability currently represented by the market. For a position S, expected settlement value is $SP(A \mid \tilde{Y}_{t_i}, I)$, while the expected acquisition cost follows from Equation 5. Equation 8 shows the expected profit as:

$$\pi(S) = S\left(\Delta_{t_i} - \frac{\lambda S}{2}\right) - c, \qquad S^* = D\Delta_{t_i}, \qquad \pi^* = \frac{D\Delta_{t_i}^2}{2} - c \tag{8}$$

where c collects the fixed costs of acting. Thus, the optimal position increases with both market depth and the actor's probability advantage, while maximum expected profit increases linearly with depth and quadratically with that advantage. This formulation covers both privileged information and manipulation. For an informed actor, I changes the estimate of $P(A)$ without changing the event itself. For example, a forecaster who has seen an unreleased ensemble may assign $P(A \mid \tilde{Y}_{t_i}, I)$ substantially above the public market price. The mathematical advantage is then the same type of probability update familiar from hydrologic forecasting, except that the information is not equally available to all participants. Manipulation is more consequential because the actor changes the settlement process rather than merely possessing a better estimate of it. If the actor can force a favorable resolution—for example, by corrupting a settling observation, interfering with an upstream data stream, or delaying a release—then the actor's effective probability of a \$1 settlement approaches unity (Equation 9). Setting:

$$P(A \mid \tilde{Y}_{t_i}, I) = 1 \tag{9}$$

and letting m denote the cost of manipulation gives Equation 10:

$$\pi^*_m = \frac{D(1 - p_{M,t_i})^2}{2} - m, \qquad D^* = \frac{2m}{(1 - p_{M,t_i})^2}, \tag{10}$$

where D^* is the breakeven market depth. Manipulation is profitable whenever $D > D^*$.

Three features of Equation 10 are important for hydrologic applications. First, the incentive grows linearly with market depth, so increasing liquidity can increase the reward for manipulating a settlement oracle. Second, the incentive grows with the square of the probability advantage. A low-probability hydrologic event trading at $p_{M,t_i} \ll 1$ therefore provides a particularly large payoff if an actor can force settlement. Third, the profit function itself does not distinguish among a scientifically superior forecast, privileged access to an unreleased forecast, and manipulation of the observation or decision on which the contract settles. All three enter mathematically through the gap between p_{M,t_i} and the actor's effective probability of settlement. The distinction among them is not mathematical but scientific, institutional, and ethical.

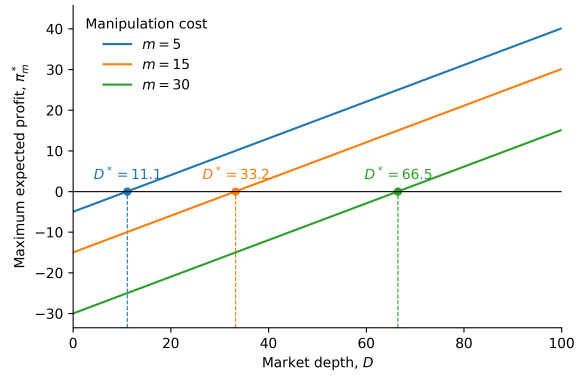


FIGURE 2
Expected profit from manipulating settlement value as a function of market depth in the stylized model, with $p_0 = 0.05$. The curves show three manipulation costs, m. Every finite manipulation cost has a breakeven depth, D^*, marked on its corresponding curve, beyond which manipulation is profitable. The axes are in arbitrary units; the figure illustrates the model and contains no empirical data.

Three conclusions follow, and Figure 2 illustrates them. First, the reward scales linearly with market depth, so thin markets deter manipulation by accident rather than by design. Every manipulation with finite cost becomes profitable at some level of liquidity, and liquidity is precisely what growing platforms cultivate. Second, writing $p_0 \equiv p_{M,t_i}$ for the pre-trade market price and p_1 for the actor's effective probability of settlement, the reward scales with the square of the informational or engineered edge, $(p_1 - p_0)^2$. Low-priced tail contracts, which are where hydrologic disasters tend to reside, therefore offer the largest potential returns to manipulation. Third, the insider case and the manipulation case have the same mathematical form, with different sources of p_1. The payoff function does not distinguish a sharper forecast from a corrupted oracle. An ethical analysis therefore cannot treat market accuracy as a sufficient defense.

The model's limitations follow directly from its assumptions. Linear price impact is a simplification. Real order books exhibit wider spreads and higher trading costs at the longshot end (Dubach, 2026), which raises c and m in the tails. Settlement rules, position limits, and market surveillance can also alter both parameters. None of these amendments, however, changes the qualitative conclusion that the incentive to manipulate grows with market depth.

9 Consequences for water management decisions

The avenues described above explain how a market can be gamed, which is a matter of values and ethics of water prediction because of what sits downstream of the manipulation. A delayed or altered forecast is more than a financial event. It is an input to operational decisions that allocate water, warn communities, and release or withhold public relief. In other words, the chain runs from mechanism to decision to the people who live with the outcome, and at no point does the affected public have visibility into why the decision was shaped as it was. This opacity is the ethical center of the problem: it fundamentally violates the principles of transparency and public consent, as communities are subjected to AI-driven resource allocations and safety decisions that harbor hidden, market-induced biases. Flood warning is the clearest case. The decisions keyed to a forecast crest (when to issue a flood watch, when to order an evacuation, and whether to pre–release a reservoir ahead of the peak) are decisions in which hours matter and timing is central. This timing is exactly what an insider position rewards. A forecaster who holds a contract that settles on a river stage benefits from widening the interval between internal knowledge and public release. In this situation, the financially optimal move is delaying the information release, a move that is operationally catastrophic. A consequentialist (utilitarian) framework exposes a fatal flaw in the pro-market defense. Even in the event that prediction markets are a utilitarian good because they theoretically yield sharper predictions that benefit society overall, when financial incentives encourage the delay of life–saving information, the actual operational consequence is maximized suffering. The resulting harm is not distributed evenly. For example, the cost of a late warning falls hardest on those least able to absorb it, such as residents of floodplain housing, households without flood insurance or a second place to go, and communities whose infrastructure offers the least margin. An incentive to delay critical information is deeply regressive, actively undermining the equitable allocation of safety resources and inflicting an invisible inequity upon vulnerable communities. Inflicting this invisible inequity transforms community survival into a tradable commodity, tearing at the social fabric by exploiting the extreme vulnerability of those in the path of the flood.

Drought declaration is the second case, and it illustrates manipulation of the settling value rather than its timing. Drought status in the United States is tied to bright-line categories, and crossing a threshold carries distributive and legal consequences, such as agricultural disaster relief, emergency water allocations, mandatory restrictions, and shifts in allocation priority among senior and junior water-rights holders. A contract that settles on whether a county enters a given drought category therefore creates a direct financial interest in where the line is drawn. Because the categorization draws on model outputs and assimilated observations, interference upstream in that data, of the kind described above, can move both the contract and the declaration together. In this way, a manipulated designation does not merely settle a bet. It redirects relief toward or away from real communities and reallocates a scarce resource under the color of objective science. The participants who gain have no stake in the allocation, and the people who depend on the allocation have no seat at the table.

These harms also corrode the decision process itself. Decision-makers and the public act on a forecast only to the degree that they trust it, and a single substantiated case of manipulation degrades that trust for every forecast that follows, which raises the real prospect that valid warnings are discounted. Specifically, alternatives generated by artificial intelligence may be accepted with little scrutiny because decision-makers believe that the model has produced the best answer. A market forming around those outputs sharpens this danger, because a manipulated or market-shaped

alternative arrives carrying both the authority of the model and the apparent confirmation of a price.

Structured decision-making and decision-making under deep uncertainty are designed to surface values, to weigh alternatives, and to act responsibly when the future is poorly known. Each of these frameworks depends on the integrity of the alternatives and probabilities that are fed into it. A market that distorts the underlying forecast, or that gives a participant a reason to delay or skew it, therefore corrupts the inputs to these frameworks even when every other step is followed faithfully. Protecting the values that these methods are designed to express therefore requires attention to how forecasts are made and to the financial incentives now accumulating around them.

10 A new conflict of interest in forecast science

Science has always managed conflicts of interest. For example, funding shapes what gets studied, industry sponsorship colors how results are framed, and the pressure to publish bends how findings are reported. These conflicts distort the production of knowledge, and the field has built disclosure norms, peer review, and replication in order to contain them. A market that settles on live forecast output is a different kind of conflict, however, and forecast science is the area in which this conflict has the greatest effect.

The difference is one of immediacy and resolvability. Most scientific output is not a tradable instrument. A result is contested, revised, and absorbed over a period of years, and there is no moment at which a single number pays out. Forecasting, however, is the exception. It produces a continuous stream of time-stamped, publicly released predictions about specific events that resolve on a known schedule and against an authoritative record. This is precisely the object that a prediction market requires, and it means that forecasters, more than other scientists, now generate output that can be bet on in real time. Whereas earlier conflicts bias what gets studied, this conflict creates a direct financial reward for getting the forecast wrong, for releasing it late, or for tilting the value on which a contract settles. This reward exists whether or not anyone acts on it. In other words, the mere presence of the market changes what it means to hold privileged access to a forecast, and it converts every preliminary report, every gauge reading, and every model run into a potential instrument of private gain.

Water forecasting concentrates this exposure. Its products resolve on short horizons against agency records, the events it predicts carry large and unevenly distributed consequences, and its data supply chain reaches through thousands of external sensors that no single party fully controls. The same features that make hydrologic forecasting valuable to the public also make it attractive to a market and vulnerable to one. The structural exposure detailed here is not unique to hydrology, as any field providing fast-resolving scientific predictions faces a similar hazard. However, water sits at the front of this problem, because its forecasts are operational and continuous and because they are tied to decisions that move money and protect lives.

10.1 Modes of engagement

It is reasonable to ask whether "participation" in these markets is ethical, as if participation were a single thing. We distinguish five modes, because the conflation invites two errors: condemning defensible conduct with the same sentence used for indefensible conduct, and letting the indefensible conduct borrow legitimacy from the defensible. Table 3 defines the modes, and the analysis in Section 10.2 is conducted per mode.

10.2 The frameworks applied

Consequentialism evaluates actions by outcomes, and the ultimate professional duty of a water manager, on this view, is to minimize suffering and save lives. The strongest consequentialist case for engagement is real and worth stating: if an adversarial, market-driven model generates a sharper tail-risk warning than internal models, ignoring a superior life-saving signal would itself be a moral failure, and protecting professional boundaries at the cost of a better warning would be the unethical choice. But the case dissolves on inspection of the modes. The benefit claimed attaches to modes 3 and 5, consuming and studying market information, and never requires modes 1 or 2, professionals holding positions. Meanwhile the oracle paradox and Goodhart dynamics (Section 4.1) mean the accuracy gain is illusory exactly where settlement drifts down the taxonomy, and Section 9 shows that the same financial incentives encourage the delay of life-saving information, so the expected consequences of professional trading are negative even before reputational harm is counted.

Deontology holds that a professional forecaster owes categorical duties of objectivity, impartiality, and public trust, and that some actions are wrong regardless of their consequences. On this view, mode 1 is a paradigm violation: it treats the public's safety and the agency's data merely as means to a financial end. An objection deserves a direct answer here. If the mere presence of financial exposure corrupted science, the argument would prove too much, since scientists own index funds, buy insurance, and live in floodplains without destroying forecast credibility. The distinction our argument requires is the one professional ethics law already draws: generalized financial exposure to an outcome differs from exposure that is specific to one's own work product and privileged access. Federal standards of conduct prohibit employees from using nonpublic government information for private gain and restrict holdings that a specific duty can affect (5 C.F.R. Part 2635; U.S. Office of Government Ethics, 2026), and the commodities fraud counts in the Van Dyke indictment show these principles reaching event contracts. For example, a forecaster who trades a contract that settles on their own forecast occupies the position of a referee betting on a match they officiate, while a forecaster who owns a diversified fund occupies the position of any citizen with savings, and professional ethics has never had difficulty distinguishing the two. The claim defended here is thus the narrow one, and it does not initiate a slope, because it follows a line that existing law has drawn for decades and applies that line to a new instrument.

TABLE 3 Modes of professional engagement with forecast-settled markets and the assessment defended in Section 10.2.

Mode	Description	Assessment
1. Trading on privileged access	A forecaster or agency employee trades contracts that settle on products they produce, influence, or see before release.	Impermissible without qualification under every framework applied here; existing law and ethics rules treat close analogues as offenses (Section 10.2).
2. Personal trading without privileged access	A professional trades water or weather contracts outside their operational responsibility, using public information and professional skill.	Impermissible as a professional norm: outsiders cannot distinguish Mode 2 from Mode 1, appearance costs are collective, and normalized professional presence in the market feeds the Goodhart dynamic.
3. Agency consumption of market signals	An agency reads market-implied probabilities as one input among many, without trading.	Conditionally permissible with safeguards, including documented provenance, statistical debiasing, and no feedback into products that settle contracts. It is of limited value where the agency is itself the oracle, since the market then partly reflects the agency back at itself.
4. Agency as involuntary settlement oracle	A platform designates agency products as the resolution source without agency consent.	Not a choice, and therefore not a locus of agency wrongdoing. It nevertheless generates defensive duties, including pipeline hardening, release protocols, and disclosure (Section 12).
5. Sanctioned adversarial study	Researchers or agencies study market strategies, probe their own pipelines, and monitor markets without holding positions.	Permissible and advisable; captures the audit benefit without the participation costs (Section 11).

Satz's (Satz, 2010) account of noxious markets supplies the third test, and it applies to the market itself rather than to any participant's conduct. A market that settles on catastrophic flood or drought relies on the desperate vulnerability of those in the event's path and converts community survival into speculative entertainment, scoring high on the dimensions of vulnerability and harm that Satz identifies. This is why mode 2 fails as a professional norm even though it involves no privileged access: professionals lending their standing to a noxious market legitimate it, and the public cannot audit which mode a professional's position occupies.

Synthesizing across frameworks and modes: trading by professionals (modes 1 and 2) is unethical, in the first case categorically and in the second as a binding professional norm; agency consumption (mode 3) is conditionally permissible but weak in value where the agency is the oracle; mode 4 generates defensive duties rather than blame; and mode 5 is where the genuine benefits live. The marginal and unproven gains to forecast accuracy cannot license the moral hazard and epistemic corruption that professional trading introduces, and the prohibition forfeits little, because the benefits at issue remain available through modes 3 and 5.

11 Capturing the audit benefit without participation

The same dynamics that create risk also create an opening. Markets reward decentralized experimentation, and the financial incentive to forecast water outcomes accurately will draw talent and computational resources toward hydrologic prediction at a scale that public budgets cannot fund. Because the physical, AI, and market forecasting workflows share partially analogous information-processing features (Table 1), this financial lure could inevitably accelerate AI advancement for water resources management. In the race to capture asymmetric upside, technically advanced participants will build increasingly sophisticated AI models, aggressively optimizing the assimilation of new observations and the generation of latent inputs to dominate the market's order book.

The question a fair reader asks is how agencies would discover these vulnerabilities without adversarial pressure, because internal review alone cannot be expected to find what an adversary would find. The answer is that adversarial pressure does not require real-money markets in disaster outcomes, because security practice has spent two decades building sanctioned substitutes. Three mechanisms transfer directly. The first is commissioned red-teaming: agencies can pay adversarial teams to probe observation networks, ingest pipelines, and release processes under rules of engagement, which is standard practice for exactly the class of infrastructure at issue. The second is coordinated vulnerability disclosure and bug bounty programs, which pay outside researchers for disclosed weaknesses roughly what a market would pay for exploiting them, with an established literature on their design (Malladi and Subramanian, 2020). The third is play-money and prize-based forecasting tournaments, which retain the aggregation and talent-attraction benefits of markets while removing settlement-linked payoffs, and which can be run with agency consent on agency terms; the forecasting tournament literature shows that skill identification does not require wagering (Tetlock and Gardner, 2015). Alongside these, mode 5 monitoring of real markets, including on-chain forensics of the kind demonstrated for decentralized order books (Dubach, 2026), extracts the informational residue of adversarial activity without holding positions. These mechanisms capture the utilitarian upside while incurring none of the participation costs. What they require is deliberate investment, and that is a governance choice, which is where this paper turns next.

12 Recommendations

The analysis above assigns different obligations to different actors, and this section collects them. A question raised by our own conclusion must be answered first, namely how a norm against professional trading becomes binding. No single instrument accomplishes this on its own; the prohibition becomes real as overlapping layers, each covering conduct the others miss: criminal and commodities law reaches insider conduct (the theory being tested in the Van Dyke prosecution); employment rules and federal standards of conduct reach agency personnel; professional society codes reach members whom employment rules do not; and platform rules reach accounts. The recommendations below build those layers.

Agencies should extend existing standards of conduct explicitly to event contracts that settle on agency products, so that no employee may hold positions in contracts resolving against outputs of their own agency. They should adopt release protocols that collapse the exploitable interval between internal availability and public availability of forecasts and records, including simultaneous publication and cryptographic signing and timestamping of releases, so that settlement-relevant data carries verifiable provenance. They should harden ingest pipelines against the upstream interference of Section 7, treating external sensors and public data feeds as untrusted inputs, and they should adopt disclosure protocols for suspected manipulation of settlement-relevant data, because an undisclosed incident compounds the loss of public trust when it eventually surfaces.

Professional societies in the hydrologic and atmospheric sciences should amend their codes of ethics to name trading in forecast-settled event contracts as a conflict of interest, distinguish the modes of Table 3 so that sanctioned study is protected while trading is proscribed, and provide guidance to members employed outside government, whom federal standards do not reach.

Regulators approve the contracts that create these exposures on regulated venues, and listing standards are the natural control point there; decentralized platforms sit outside listing approval, which makes the platform obligations below load-bearing rather than supplementary. Designation of an agency product as a settlement source should require notice to, and ideally consent of, the agency named. Contracts referencing government products should carry insider provisions comparable to those governing securities, and referral channels between platforms and enforcement agencies, which functioned in the Van Dyke case, should be formalized rather than improvised.

Platforms decide what is tradable, and their choices are not neutral. Due diligence on settlement sources, published resolution criteria fixed before listing, position transparency sufficient for outside audit, and account restrictions for employees of named source agencies are all within platform control today. The record shows platforms adopting such rules only after incidents, and adopting them before disaster contracts are listed would move platform governance from remediation to prevention.

Affected communities are the parties every other actor claims to serve, and they currently have no standing in the system at all. At minimum, the listing of a contract that prices a specific community's disaster should trigger notice to that community's government, and market data on such contracts should be available to researchers and affected jurisdictions rather than held as proprietary data. Whether communities should hold a veto is a question we leave open, but the present arrangement, in which their survival can be listed without their knowledge, is difficult to reconcile with any account of consent.

13 Conclusion and discussion

The integration of values and ethics into AI-driven water management has so far focused on bias in training data, consent and ownership in data acquisition, and equity in resource allocation. These concerns remain central. We have argued, however, that they do not exhaust the problem. A financial layer is forming around forecast products, one in which artificial intelligence lowers the cost of probabilistic prediction and prediction markets convert those forecasts into tradable and automatically settled contracts. When an agency forecast becomes the oracle for a real-money market, the incentive to game that market reaches back into the data, the model, and the operational decisions that allocate water and warn communities. In this way, the values that ought to govern a forecast (accuracy, impartiality, and public benefit) come into direct tension with the values of a market, which reward profit regardless of how that profit is made.

Our conclusions are narrower and, we believe, stronger than a blanket verdict on participation. Professional trading in forecast-settled contracts is unethical: categorically where privileged access is involved, and as a binding professional norm even where it is not, because the public cannot audit the difference and the profession's credibility is a shared asset. Agency consumption of market signals is at most conditionally permissible and is weakest exactly where the agency is the settlement oracle, since the market then partly reflects the agency back at itself. The genuine benefits these markets promise, adversarial discovery of vulnerabilities and the attraction of talent to hydrologic prediction, are available through sanctioned red-teaming, disclosure programs, and non-wagering tournaments, and thus prohibiting professional trading imposes little cost on the public.

The cost of that tension is paid in credibility, and credibility is the asset that forecast science can least afford to lose. A market that settles on live forecasts gives individuals a financial reason to corrupt the very output that the public must trust, and the harm of any single case is borne by every honest forecaster whose warnings are discounted afterward. The stylized model of Section 8 adds a warning about timing: thin markets deter manipulation only by accident, and the protection erodes as liquidity grows, so the window for establishing rules is closing. This is a new source of incentive to act unethically, and it is sharpest where forecasts are operational and carry heavy consequences, which is to say that it is sharpest in water. The forecasting community should therefore decide, deliberately and soon, whether its products should function as financial oracles, and on what terms, before the market decides on its behalf. Because this structural vulnerability is not unique to hydrology, we call upon experts in other environmental hazard forecasting disciplines (such as seismology, epidemiology, and wildfire management) to analyze the prediction market ethics of their respective fields before these platforms become fully entrenched.

Author contributions

TL: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. JF: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that Generative AI was used in the creation of this manuscript. Both authors used generative AI tools in the preparation and revision of this manuscript and take full responsibility for its content. The tools used were Claude (Anthropic) and ChatGPT (OpenAI). They were used to revise, to check arguments for internal consistency, and to help identify literature that the author(s) then retrieved and verified against the primary sources. No generative AI tool was used to fabricate or analyze data, to generate or alter any figure or table, or to substitute for the author(s)' analytical and interpretive judgment. The author(s) are solely responsible for the accuracy, originality, and integrity of the final work. No AI tool is credited as an author.

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